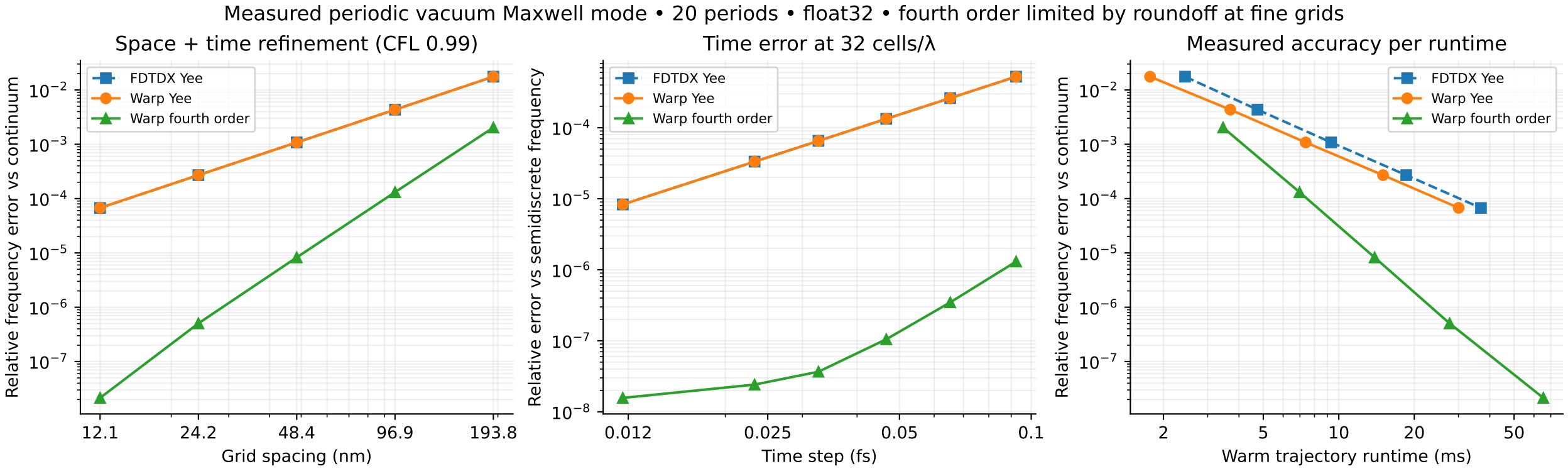
Measured periodic vacuum Maxwell mode • 20 periods • float32 • fourth order limited by roundoff at fine grids
Space + time refinement (CFL 0.99)
Time error at 32 cells/λ
Measured accuracy per runtime
FDTDX Yee
Warp Yee
Warp fourth order
Relative frequency error vs continuum
Relative error vs semidiscrete frequency
Grid spacing (nm)
Time step (fs)
Warm trajectory runtime (ms)
12.1
24.2
48.4
96.9
193.8
0.012
0.025
0.05
0.1
2
5
10
20
50